wtw158 **TECHNIQUE MASTERING QUESTIONS**

SECTION I DIFFERENTIATION

Some of the answers are simplified. We need the simplified version when we want to sketch graphs and in the textbook the answers are mostly simplified.
<u>**You DO NOT have to simplify your answers in this exercise.**</u>

1. $8x^3 - 6x + 5$

2. $\frac{-1}{2}x^{\frac{-3}{2}} + \frac{1}{2}x^{\frac{-1}{2}} = \dfrac{x-1}{2x\sqrt{x}}$

3. $6x + (-2)\frac{1}{3}x^{-3} + 1 = \dfrac{18x^4 - 2 + 3x^3}{3x^3}$

4. $2x - 3\left(\frac{7}{3}\right)x^{\frac{4}{3}} = 2x - 7x^{\frac{4}{3}}$

5. $1 + 0 - 2t^{-3} = \frac{t^3 - 2}{t^3}$

6. $\frac{1}{2}x^{\frac{-1}{2}} - \left(-\frac{1}{2}\right)x^{\frac{-3}{2}} = \dfrac{x+1}{2x\sqrt{x}}$

7. $12(-1)x^{-2} - 4(-3)x^{-4} + (-4)x^{-5} = \dfrac{-4(3x^3 - 3x + 1)}{x^5}$

8. $\frac{1}{3}x^{\frac{-2}{3}} - \frac{3}{x^4}$

9. $4x^3 - 8(3)x^2 + 2(2)x - 1 + 0 = 4x^3 - 24x^2 + 4x - 1$

10. $7(3)y^2 - 5(2)y + 3 + 0 = 21y^2 - 10y + 3$

11. $9(-3)x^{-4} + 2(-2)x^{-3} + 0 = -\dfrac{27 + 4x}{x^4}$

12. $-2(4)x^3 + (-2)x^{-3} - 3(\frac{3}{4})x^{\frac{-1}{4}} = --\dfrac{32x^{\frac{25}{4}} + 8\sqrt[4]{x} + 9x^3}{4x^{\frac{13}{4}}}$

13. $12(4)t^3 + 3(3)t^2 + 5(-2)t^{-3} - 0 = \dfrac{48t^6 + 9t^5 - 10}{t^3}$

14. $4(-2)x^{-3} - 7(\frac{1}{2})x^{\frac{-1}{2}} + 8(3)x^2 + 0 = -\frac{16\sqrt{x} + 7x^3 - 48x^{\frac{11}{2}}}{2x^{\frac{7}{2}}}$

15. $3(3)x^2 = 2(2)x - 1 + 0 = 9x^2 + 4x - 1$

16. $4(3)x^2 - 7(2)x + 8 - 0 = 12x^2 - 14x + 8$

17. $3x^2 - 3 - 2(-4)x^{-5} = 3x^2 - 3 + 8x^{-5}$

18. $1 + 0 + 4(-1)x^{-2} = 1 - \dfrac{4}{x^2}$

19. $-3(-3)u^{-4} + 4(2)u + (-2)u^{-3} = 9u^{-4} + 8u + \frac{-2}{u^3}$

20. $\frac{d}{dx}\left(x^{\frac{3}{2}} + x^{\frac{-5}{2}}\right) = \dfrac{3x^4 - 5}{2x^{\frac{7}{2}}}$

21. $5(x^3 + 7x^2 + 5)^4(3x^2 + 7(2)x + 0) = 5(x^3 + 7x^2 + 5)^4(3x^2 + 14x)$

22. $\frac{1}{2}(x^4+x^2+2)^{\frac{-1}{2}}(4x^3+2x+0) = \frac{1}{2}(x^4+x^2+2)^{\frac{-1}{2}}(4x^3+2x)$

23. $\frac{1}{2}(x+1)^{\frac{-1}{2}}(x^2+1) + \sqrt{x+1}\,)(2x) = \dfrac{5x^2+4x+1}{2\sqrt{1+x}}$

24. $\frac{1}{2}\left(t^2+2t+3\right)^{\frac{-1}{2}}\left(2t+2\right) = \dfrac{t+1}{\sqrt{t^2+2t+3}}$

25. $2x(x^3-1)+x^2(3x^2) = 5x^4-2x$

26. $4(x^3-3x)^3(3x^2-3) = 12x(x^2-3)(x^2-1)$

27. $\sqrt{1-x^2} + (x)\frac{1}{2}(1-x^2)^{\frac{-1}{2}}(-2x) = \dfrac{1-2x^2}{\sqrt{1-x^2}}$

28. $(2)(3x^2+2)+(2x-4)(6x) = 18x^2-24x+4$

29. $[3(2x-3)^2(2)](4x+2)^2+(2x-3)^3[(2)(4)(4x+2)]$

30. $[2(7x+3)(7)](3x^2-14x+5)+(7x+3)^2(3(2)x-14)$

31. $(24y^2)(3y^2-5y+10)^2+(8y^3-2)[2(3y^2-5y+10)(6y-5)]$

32. $\frac{2}{3}(2x^3-3)^{\frac{-1}{3}}(6x^2)$

33. $\frac{1}{2}(3x^2-2x+1)^{\frac{-1}{2}}(6x-2) = \dfrac{3x-1}{\sqrt{3x^2-2x+1}}$

34. $2(3x+4)+(2x-3)(3) = 12x-1$

35. $(6x^2)(x^4+x)+(2x^3-1)(4x^3+1) = 14x^6+4x^3-1$

36. $(\frac{-1}{x^2})(x^2-5)+(\frac{1}{x}+3)(2x) = \dfrac{6x^3+x^2+5}{x^2}$

37. $4(2x+1)(x^2+2)^3+(2x+1)^2 3(x^2+2)^2(2x) = 2(2x+1)(x^2+2)^2[8x^2+3x+2]$

38. $\frac{1}{2}(3x^2+1)^{\frac{-1}{2}}(6x) = \dfrac{3x}{\sqrt{3x^2+1}}$

39. $6(2t-5)^2$

40. $-3(6)(6x-5)^{-4} = -\dfrac{18}{(6x-5)^4}$

41. $\frac{(2x+1)(x^2+1)-(x^2+x+1)2x}{(x^2+1)^2} = \dfrac{1-x^2}{(x^2+1)^2}$

42. $\frac{2(3x+2)-(2x+3)(3)}{(3x+2)^2} = \dfrac{-5}{(3x+2)^2}$

43. $\frac{2(3u-5)-(2u+1)(3)}{(3u-5)^2} = -\dfrac{13}{(3u-5)^2}$

44. $\frac{(2x+5)x^2-(x^2+5x-1)(2x)}{(x^2)^2} = \dfrac{2-5x}{x^3}$

45. $\frac{d}{dx}\left(\dfrac{(x^2+3x+2)}{(x^2-3x+2)}\right) = \dfrac{(2x+3)(x^2-3x+2)-(x^2+3x+2)(2x-3)}{(x^2-3x+2)^2} = \dfrac{-6(x^2-2)}{(x-1)^2(x-2)^2}$

46. $\frac{1}{2}\left(\frac{x^2+1}{x^2+4}\right)^{\frac{-1}{2}}\left(\frac{(2x)(x^2+4)-(x^2+1)2x}{(x^2+4)^2}\right) = \frac{3x}{\sqrt{x^2+1}\,\sqrt{x^2+4}\,(x^2+4)^2}$

47. $\frac{1}{2\sqrt{t}}(1+5t^2) = \frac{(1+3t^2)\sqrt{t} - (t+t^3)\frac{1}{2}t^{\frac{-1}{2}}}{(\sqrt{t})^2} = \frac{1+5t^2}{2\sqrt{t}}$

48. $2\left(\frac{x+1}{x-1}\right)\left(\frac{(x-1)-(x+1)}{(x-1)^2}\right) = -\frac{-4(x+1)}{(x-1)^3}$

49. $\frac{d}{dx}\left(10(\sqrt{x}+4)^{-1}\right) = -10(\sqrt{x}+4)^{-2}\left(\frac{1}{2}x^{\frac{-1}{2}}\right) = \frac{-5}{(\sqrt{x}+4)^2\sqrt{x}}$

50. $\frac{d}{dx}\left(8(4+x^2)^{-1}\right) = -8(4+x^2)^{-2}(2x) = -\frac{16x}{(4+x^2)^2}$

51. $\frac{4(y^2+1)-4y(2y)}{(y^2+1)^2} = \frac{4(1-y^2)}{(y^2+1)^2}$

52. $\frac{d}{dx}\left(3(2x+1)^{\frac{-1}{2}}\right) = 3(\frac{-1}{2})(2x+1)^{\frac{-3}{2}}(2) = -\frac{3}{(2x+1)^{\frac{3}{2}}}$

53. $\frac{(6x-2)(4x^2-5)-(3x^2-2x+3)(8x)}{(4x^2-5)^2} = \frac{2(4x^2-27x+5)}{(4x^2-5)^2}$

54. $\frac{[(2)(3x+5)+(2x-4)(3)](2x^2+7)-(2x-4)(3x+5)(4x)}{(2x^2+7)^2} = \frac{2(82x+2x^2-7)}{(2x^2+7)^2}$

55. $\frac{2(x^2+2x)-(2x-3)(2x+2)}{(x^2+2x)^2} = \frac{-2(x^2-3x-3)}{x^2(x+2)^2}$

56. $\frac{(-2+6t)(t^2+2)-(4-2t+3t^2)2t}{(t^2+2)^2} = \frac{2(t^2+2t-2)}{(t^2+2)^2}$

57. $6x^2 + \frac{-1(x^3)-(2-x)(3x^2)}{x^6} = \frac{2(3x^6+x-3)}{x^4}$

58. $\frac{1(x+1)-(x-1)1}{(x+1)^2} = \frac{2}{(x+1)^2}$

59. $\frac{2x(x^2+1)-x^2(2x)}{(x^2+1)^2} = \frac{2x}{(x^2+1)^2}$

60. $\frac{d}{dt}(t^4-2t+1)^{-1} = -1(t^4-2t+1)^{-2}(4t^3-2) = \frac{-2(2t^3-1)}{(t^4-2t+1)^2}$

61. $\frac{d}{dx}(\sin(x^2+2)) = [\cos(x^2+2)](2x) = 2x\cos(x^2+2)$

62. $(\cos(3x))(3) = 3\cos 3x$

63. $(-\sin(3y))(3) = -3\sin 3y$

64. $[-\sin(x^4+7)]4x = -4x^3\sin(x^4+7)$

65. $[\sec^2(\sin x)]\cos x = \cos x \times \sec^2(\sin x)$

66. $(\sec^2(t^2+5))(2t) = \frac{2t}{\cos^2(t^2+5)}$

67. $[\cos(3x+2)](3) = 3\cos(3x+2)$

68. $2\cos x - \sec^2 x = \dfrac{2\cos^3 x - 1}{\cos^2 x}$

69. $= 1 + \sin x$

70. $\dfrac{(\cos x)x - \sin x(1)}{x^2} = \dfrac{x\cos x - \sin x}{x^2}$

71. $[\sec^2(2x - x^3)](2 - 3x^2) = \dfrac{2 - 3x^2}{\cos^2(-2x + x^3)}$

72. $(\cos(3\theta^2 - 2\theta + 1))(6\theta - 2) = (6\theta - 2)\cos(3\theta^2 - 2\theta + 1)$

73. $3[\cos(x^2)](2x) + 2\cos x - 0 = 6x\cos x^2 + 2\cos x$

74. $3\cos x + 2\sin x$

75. $(-\sin(\pi x - 1))(\pi) = -\pi\sin(\pi x - 1)$

76. $3[-\sin(3x)] - 3\sec^2(3x) = \dfrac{-3(\sin 3x\cos^2 3x + 1)}{\cos^2 3x}$

77. $2x\sin x + x^2\cos x = x(2\sin x + x\cos x)$

78. $[-\sin(\tan x)]\sec^2 x = \dfrac{-\sin(\tan x)}{\cos^2 x}$

79. $2\cos(2x+3)$

80. $(\cos 2\pi t)(2\pi) = 2\pi(\cos 2\pi t)$

81. $\left(\cos\frac{1}{\sqrt{x}}\right)\left(-\frac{1}{2}x^{\frac{-3}{2}}\right) = -\dfrac{\cos\frac{1}{\sqrt{x}}}{2x^{\frac{3}{2}}}$

82. $(\cos\frac{1}{x})\left(-\frac{1}{x^2}\right) = -\dfrac{\cos\frac{1}{x}}{x^2}$

83. $(-\sin(-x))(-1) = -\sin x$

84. $\left(\sec^2\pi(\frac{1}{2} - x)\right)(-\pi)$

85. $2x\sec^2(x^2+1)$

86. $5\sec^2 5t$

87. $\cos x + x(-\sin x) = \cos x - x\sin x$

88. $\dfrac{(\cos x)(1+x) - \sin x}{(1+x)^2}$

89. $4(2)(\sin 3x)(\cos 3x)(3) = 12\sin 6x$

90. $\dfrac{(-\sin x)\sin x - \cos x(\cos x)}{(\sin x)^2} = \frac{-1}{(\sin x)^2}$

91. $2x\sin x + (x^2+3)\cos x$

92. $\frac{1}{2}(\sin x)^{-\frac{1}{2}}\cos x = \dfrac{\cos x}{2\sqrt{\sin x}}$

93. $2(\cos(x^3))(-\sin(x^3))(3x^2) = -3x^2(\sin 2x^3)$

94. $\cos 5x + x(-\sin 5x)(5) = \cos 5x - 5x\sin 5x$

95. $2\cos(2y) \times \cos(3y) + \sin(2y)(3(-\sin(3y))) = 2\cos 2y\cos 3y - 3\sin 2y\sin 3y$

96. $2(\sin(3x))(3)\cos(3x) + \cos(5x)(2)(-\sin(5x))(5) = 3\sin 6x - 5\sin 10x$

97. $2x\tan\frac{1}{x} + x^2(\sec^2(\frac{1}{x}))\left[\frac{-1}{x^2}\right] = 2x\tan\frac{1}{x} - \sec^2(\frac{1}{x})$

98. $-1(\sin x - x\cos x)^{-2}[\cos x - (\cos x + x(-\sin x))] = \dfrac{-x\sin x}{(\sin x - x\cos x)^2}$

99. $\dfrac{(\cos x - \sin x)(\tan x) - (\sin x + \cos x)\sec^2 x}{(\tan x)^2}$

100. $\frac{d}{dx}(1) = 0$

101. $\dfrac{2x\tan x - x^2(\sec^2 x)}{\tan^2 x}$

102. $\frac{1}{2}(\cos(2x))^{\frac{-1}{2}}(-\sin(2x))(2) = \dfrac{-\sin 2x}{\sqrt{\cos 2x}}$

103. $2\sin(x-3)\cos(x-3) = \sin(2(x-3))$

104. $3\cos^2(t^2-t)[-\sin(t^2-t)](2t-1) = -3(2t-1)\cos^2(t^2-t)\sin(t^2-t)$

105. $\cos x\cos^2 x + (\sin x)2(\cos x)(-\sin x) = \cos x(\cos^2 x - 2\sin^2 x)$

106. $\frac{1}{3}\left[\sin^{\frac{-2}{3}}(2x)\right][\cos(2x)](2) = \dfrac{2\cos 2x}{3\sin^{\frac{2}{3}} 2x}$

107. $\left[7\sin^6(2-4x)\right](\cos(2-4x))(-4) = -28\sin^6(2-4x)\cos(2-4x)$

108. $2\cos x + 6\sin x\cos x = 2\cos x(1 + 3\sin x)$

109. $3\cos^2 u(-\sin u) + 4(2)\cos u(-\sin u) = -\sin u\cos u(3\cos u + 8)$

110. $5(2)[\cos(x+2)](-\sin(x+2)) + 3(-\sin(x+2)) = -\sin(x+2)(10\cos(x+2) + 3)$

111. $10x - 6\tan x\sec^2 x + \cos x = \dfrac{10x\cos^3 x - 6\sin x + \cos^4 x}{\cos^3 x}$

112. $\dfrac{(\sec^2 x)(1-\tan x) - (\tan x)\left(-\sec^2 x\right)}{(1-\tan x)^2}$

113. $\frac{3}{2}\sin^{\frac{1}{2}}(x+1)\cos(x+1) = \frac{3}{2}\sin^{\frac{1}{2}}(x+1)\cos(x+1)$

114. $\frac{d}{dx}(1) = 0$

115. $(\cos\sqrt{x})\frac{1}{2}x^{\frac{-1}{2}} + \frac{1}{2}(\sin x)^{\frac{-1}{2}}\cos x = \dfrac{(\cos\sqrt{x})}{2\sqrt{x}} + \dfrac{\cos x}{2\sqrt{\sin x}}$

116. $(\cos x)(\sin x + \cos x) + (\sin x)(\cos x - \sin x) = \sin 2x + \cos 2x$

117. $\frac{d}{dx}(\sin 2x) = 2\cos 2x = 2\cos 2x$

118. $\frac{d}{du}(\sin u)^{-1} = (-1)(\sin u)^{-2}\cos u$

119. $2(\cos x)(-\sin x)(\sin x) + \cos^2 x\cos x = 3\cos^3 x - 2\cos x$

120. $\frac{d}{dx}\sqrt{\cos^2 x} = \frac{1}{2}(\cos^2 x)^{\frac{-1}{2}}(-2\sin x\cos x)$

121. $1 + 2\tan x\sec^2 x$

122. $\dfrac{\cos x + x\sin x}{\cos^2 x}$

123. $\sec^2 x\cos x + \tan x(-\sin x) = \cos x$

124. $\cos t\cos t + \sin t(-\sin t) = \cos^2 t - \sin^2 t = \cos 2t$

125. $2xe^{x^2}$

126. $5e^{5x}$

127. $(2x+3)e^x + (x^2+3x)e^x = e^x(x^2+5x+3)$

128. $e^x + xe^x - (-1)e^{-x} = \dfrac{e^{2x} + xe^{2x} + 1}{e^x}$

129. $\frac{d}{dx}\left(e^{x^2+x+1}\right) = (2x+1)e^{x^2+x+1} = (2x+1)e^{x^2+x+1}$

130. $\frac{d}{dx}\left(e^{x^2-x+1}\right) = (2x-1)e^{x^2-x+1} = (2x-1)e^{x^2-x+1}$

131. $-e^{-u}$

132. $\frac{2}{3}e^{\frac{2x}{3}}$

133. $e^{\sqrt{x}}\,\frac{1}{2}x^{\frac{-1}{2}} = \dfrac{e^{\sqrt{x}}}{2\sqrt{x}}$

134. $3e^{3x} + 2(2)e^{2x} - 3e^x = 3e^{3x} + 4e^{2x} - 3e^x$

135. $2xe^{x^2-2}$

136. $\dfrac{2e^{2x}(2-e^{2x}) - (1+e^{2x})(-2e^{2x})}{(2-e^{2x})^2} = \dfrac{6e^{2x}}{(2-e^{2x})^2}$

137. $3e^{3x-1} - 4(-1)e^{-x} = \dfrac{3e^{4x-1}+4}{e^x}$

138. $(-\sin(e^t))e^t = -e^t\sin(e^t)$

139. $3(2)e^{2x} - 4e^x = 2e^x(3e^x - 2)$

140. $e^{3\cos 2x} \times 3(-\sin 2x)(2) = -6(\sin 2x)e^{3\cos 2x}$

141. $-2e^{-2x} + 4(-3)e^{-3x} = -2e^{-2x} - 12e^{-3x}$

142. $2e^{2x+1}$

143. $\frac{1}{2}(2)e^{2x} = e^{2x}$

144. $(\cos x)e^{\sin x}$

145. $2e^{2x}$

146. $2e^t + 2te^t = 2e^t(1+t)$

147. $\frac{d}{dx}(1-e^{-x})^{-1} = -1(1-e^{-x})^{-2}(-(-1)e^{-x}) = \dfrac{-e^{-x}}{(1-e^{-x})^2}$

148. $\dfrac{-e^{-x}x - e^{-x}(1)}{x^2} = \dfrac{-e^{-x}(x+1)}{x^2}$

149. $2xe^{-x} + x^2(-1)e^{-x} = \dfrac{x(2-x)}{e^x}$

150. $e^{-\frac{1}{x^2}}(-1(-2)x^{-3}) = \frac{2}{x^3}e^{-\frac{1}{x^2}}$

151. $e^{\sqrt{x^2+1}}\left(\frac{1}{2}(x^2+1)(2x)\right) = \dfrac{xe^{\sqrt{x^2+1}}}{\sqrt{x^2+1}}$

152. $\frac{e^x+e^{-x}}{2}$

153. $2e^{2x}\cos 3x + e^{2x}(3)(-\sin 3x) = e^{2x}(2\cos 3x - 3\sin 3x)$

154. $4(-\sin 4x)e^{\cos 4x} = -4\sin 4xe^{\cos 4x}$

155. $(2x)2^x + x^2(2^x\ln 2) = 2^x(2x + (\ln 2)x^2)$

156. $3^{5x}5\ln 3 = (5\ln 3)3^{5x}$

157. $4x^3 + 4^x\ln 4 = 4x^3 + 4^x\ln 4$

158. $9^{-x}(-1)\ln 9 = \dfrac{-\ln 9}{9^x}$

159. $(\sec^2 5^x)(5^x\ln 5) = \dfrac{5^x\ln 5}{\cos^2 5^x}$

160. $3^{4x+1}(4\ln 3) + 2^{4x+2}(4\ln 2) = 4[(\ln 3)3^{4x+1} + (\ln 2)2^{4x+2}]$

161. $3^{y^2+1}(2y\ln 3) = (2y\ln 3)3^{y^2+1}$

162. $2^x\ln 2 = (\ln 2)2^x$

163. $2^{-x}(-1)\ln 2 = \frac{-\ln 2}{2^x}$

164. $(\frac{1}{2})^t\ln(\frac{1}{2}) = -\ln 2(\frac{1}{2})^t$

165. $e^x\ln x + e^x(\frac{1}{x}) = e^x(\ln x + \frac{1}{x})$

166. $\dfrac{\cos x}{\sin x} = \cot x$

167. $\frac{d}{dx}(\ln x)^{-1} = -1(\ln x)^{-2}(\frac{1}{x}) = \frac{-1}{x(\ln^2 x)}$

168. $\frac{d}{dx}(\ln 3x + \ln e^{-x}) = \frac{d}{dx}(\ln 3x - x) = \dfrac{3}{3x} - 1 = \frac{1}{x} - 1$

169. $\frac{d}{dx}(\ln(x-1) - \ln(x^2+1)) = \dfrac{1}{x-1} - \dfrac{2x}{(x^2+1)} = \dfrac{-(x^2-2x-1)}{(x-1)(x^2+1)}$

170. $\frac{d}{dx}(\ln e^x - \ln(1+e^x)) = \frac{d}{dx}(x - \ln(1+e^x)) = 1 - \dfrac{e^x}{1+e^x} = \dfrac{1}{1+e^x}$

171. $\frac{d}{dx}(\sin 2x) = 2\cos 2x$

172. $\frac{d}{dx}\left(\frac{1}{2}[\ln x - \ln(x^2+1)]\right) = \frac{1}{2}\left[\dfrac{1}{x} - \dfrac{2x}{x^2+1}\right] = \dfrac{1-x^2}{2x(x^2+1)}$

173. $\frac{d}{du}(2\ln u) = \dfrac{2}{u}$

174. $\frac{d}{dx}(\ln 10 - \ln x) = -\dfrac{1}{x}$

175. $\frac{d}{dx}(x\ln 10) = \ln 10$

176. $\frac{3}{3x} + \frac{4}{x} = \frac{5}{x}$

177. $2x\ln 2x + x^2\left(\frac{2}{2x}\right) = x(\ln 4 + 2\ln x + 1)$

178. $\frac{d}{dx}(-\ln x) = -\frac{1}{x}$

179. $\ln x + x(\frac{1}{x}) = \ln x + 1$

180. $-\frac{1}{x}$

181. $3(\ln y)^2\frac{1}{y} = \frac{3\ln^2 y}{y}$

182. $\ln\sqrt{x} + x\frac{d}{dx}\left(\frac{1}{2}\ln x\right) = \ln\sqrt{x} + x\left(\frac{1}{2}\right)(\frac{1}{x}) = \ln\sqrt{x} + \frac{1}{2}$

183. $\frac{7}{7x}$

184. $\frac{1}{2}(\ln x)^{\frac{-1}{2}}(\frac{1}{x}) = \frac{1}{2x\sqrt{\ln x}}$